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Assessment Paper AY 2023 Semester 2 E211 MUE Main (1)

Section A (12 Questions)

Q1. Which one of the below statements is a **correct** assumption of a **M/M/2** queuing model?

- Number of channels is 2 (Correct Option)
- Number of service phases is 2
- Service time is constant
- Arriving population may exhibit balking behaviour

Q2. Which one of the following is **NOT** a pitfall to conducting a successful simulation study?

- Failure to properly define the objectives and scope of a simulation study
- Overly dependent on the simulation model to provide an optimal solution to the problem
- Failure to validate the simulation model before experimenting with scenarios
- Collecting adequate information about the system through observation and interviews (Correct Option)

Q3. Referring to the partial FlexSim process flow model in Figure 1a, what would happen if 'Acquire cashier' activity is removed?

Figure 1a: A Partial Process Flow Model

- The token will not proceed to 'Payment' activity
- Cashier resource will not be in use (Correct Option)
- The token will not complete 'Payment' activity
- The token will skip 'Release cashier' activity

Q4. Which of the following methods is used to solve a shortest path problem?

- Clarke and Wright Savings method
- Ford Fulkerson's algorithm
- Dijkstra's algorithm (Correct Option)
- Nearest neighbour method

Q5. Referring to the network diagram in Figure 1b, the maximum flow along path A-B-C-D is:

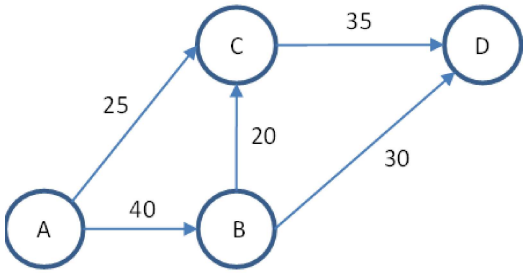

Figure 1b: A Network Diagram

- 20 (Correct Option)
- 35
- 40
- 95

Q6. Alex wants to forecast the demand for a new product. He consults the sales personnel, and they would provide him with estimates of how well the product would be received by the market. This form of forecasting is known as:

- Market research
- Consensus panel
- Delphi method
- Grassroots forecasting (Correct Option)

Q7. *Show your workings clearly.*

Q7a. A distribution network diagram is shown in Figure 2a. The nodes represent locations and numbers on the arcs represent the distances between locations.

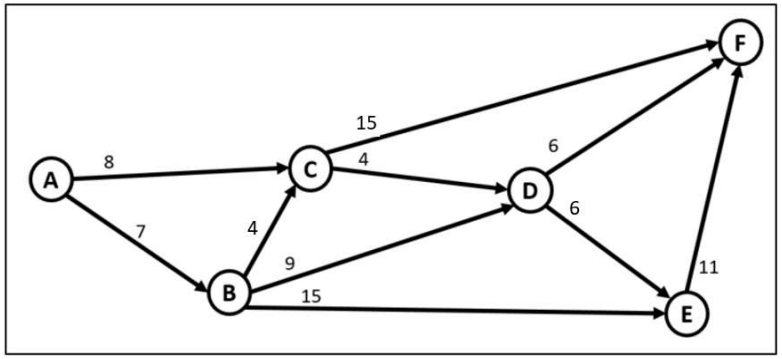

Figure 2a: A Distribution Network

Table 2a below shows **partial** workings used to find the shortest path in Figure 2a from node A to each of the other nodes.

Table 2a: Partial Workings to Solve Shortest Path Problem

Iteration	Solution Set	Arcs	Total Distance
Iteration 1	A	A → B A → C	7 8
Iteration 2	A B (7)	A → C B → C B → D B → E	8 **W** 16 22
Iteration 3	A B (7) C (8)	B → D B → E C → D **X**	16 22 12 **Y**
Iteration 4	…	…	…
…	…	…	…

Q7a(i). Calculate the value of **W** in Table 2a.

Q7a(ii). At Iteration 3, what is the arc represented by **X** and its corresponding Total Distance **Y?**

Q7a(iii). Which is the node to be added into the solution set at **Iteration 4**? Explain.

Q7a(iv). List down the shortest path from node A to node E and state the total distance.

Q7b. A Linear Programming (LP) model is formulated to solve a Minimum Cost Flow problem for the network shown in Figure 2a. Assume that the decision variable Xij represents the quantity of shipment from node i to j. (i, j = A, B, C, D, E or F).

- Supply node A can supply up to 1000 units.
- Nodes D, E and F have demands of 200, 300 and 500 units respectively.
- Nodes B and C are transshipment hubs with zero supply and demand.
- Values on arcs represent unit cost of shipment.

Construct the below constraints based on the decision variables defined.

Q7b(i). Net flow constraint for Node C.

Q7b(ii). Net flow constraint for Node E.

Q7b(iii). Maximum shipping capacity along Arc B-D is 250.

Q7c. In one city, a product has to be distributed from a warehouse (O) to a number of stores (A, B, C, D, E). Table 2b shows the distance between each pair of locations.

Table 2b: Travel Distance Between Locations

From	To (Travel distance in kilometers)

	O	A	B	C	D	E
O	-	31	19	35	33	34
A	31	-	13	21	23	6
B	19	13	-	24	26	25
C	35	21	24	-	15	18
D	33	23	26	15	-	25
E	34	6	25	18	25	-

Clark and Wright savings method is used to plan a delivery route. Table 2c shows the **savings** in distance between each pair of stores ranked in descending order.

Table 2c: Savings Ranked in Descending Order

Rank	Savings	From	To
1	59	A	E
2	53	C	D
3	51	C	E
4	45	A	C
5	42	D	E
6	41	A	D
7	37	A	B
8	?	B	C
9	28	B	E
10	26	B	D

Q7c(i). Calculate the Savings between nodes B and C.

Q7c(ii). Based on the Savings table in Table 2c, determine the delivery route using Clark and Wright savings Method. Show workings by writing down the partial tours and the rank at which each tour is constructed. State the total distance for the route.

Q7c(iii). What is another method can be used for planning delivery routes? Will any of the 2 methods guarantee an optimal solution to routing problems?

Q8. *Show your workings clearly and round off your answer to **3 decimal places** where necessary.*

Kelvin works in a company that develops high-tech gadgets. His role in the company is sales and demand planning.

Q8a. The company launched a fitness tracker watch some months ago. To predict customer demand, Kelvin collected past sales data (Table 3a) and carried out a forecast using moving average method.

Table 3a: Historical Sales from Past 6 Months

Month	Sales Figure
1	86
2	98
3	79
4	104
5	82
6	96

Q8a(i). Calculate the forecast for Month 7 based on a **3-month** moving average.

Q8a(ii). Calculate the forecast for Month 7 based on a **3-month weighted** moving average using weights of 3 (most recent), 2, 1 (least recent).

Q8a(iii). Which of the above forecast methods is more responsive to changes in the sales data?

Q8b. For another product, Kelvin collected its quarterly sales figures for the past 3 years (Figure 3a). Should Kelvin use Moving Average method to forecast demand for this product? Give a reason.

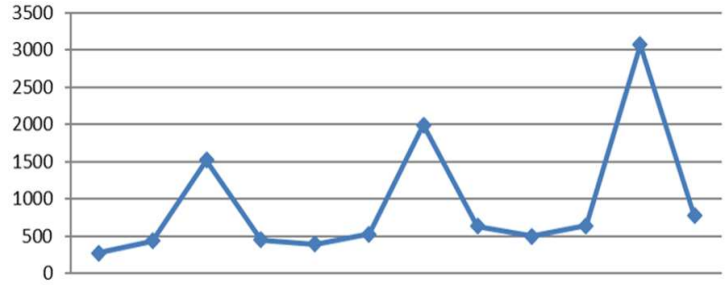

Figure 3a: Quarterly Sales Chart

Q8c. Kelvin developed an Excel template (Table 3b) to help him do the demand forecasting for a 3rd product.

Table 3b: Partial Workings of a Forecasting Method

Input Data			Seasonal Index Computation					Forecast	
Month	Actual Value (Y)	Time Period (X)	Moving Average	Centred Moving Average	Seasonal Ratio	Seasonal Index	Deseasonalized value	Deseasonalized forecast	Seasonalized forecast
2018Q1	432	1				0.843	512.330	560.962	473.007
2018Q2	564	2	682			0.894	630.564	634.891	567.870
2018Q3	656	3	**(A)**		0.915	0.935	701.910	708.820	662.458
2018Q4	1076	4		786	**(C)**	1.328	810.385	782.749	1039.307
2019Q1	710	5	904	862.25	0.823	0.843	**(E)**	856.679	722.357
2019Q2		6	992.5	948.25	0.886	0.894	939.138	930.608	832.370
2019Q3	990	7	1068.5	**(B)**	0.961	0.935	1059.285	1004.537	938.832
2019Q4		8	1138	1103.25	1.296	1.328	1076.998	1078.466	1431.948
2020Q1	1014	9	1195.5	1166.75	0.869	**(D)**	1202.553	1152.395	971.707
2020Q2		10	1263	1229.25	0.909	0.894	1249.948	1226.324	1096.870
2020Q3	1220	11				0.935	1305.382	1300.253	1215.207
2020Q4	1700	12				1.328	1280.348	1374.182	1824.590
				Intercept for 'Deseasonalized Value'			**487.033**		
				Slope for 'Deseasonalized Value'			**73.929**		

Q8c(i). How many periods of data are used to calculate the values in the Moving Average column?

Q8c(ii). Calculate the Moving Average labelled as **(A)** in Table 3b.

Q8c(iii). Calculate the Centred Moving Average labelled as **(B)** in Table 3b.

Q8c(iv). Calculate the Seasonal Ratio labelled as **(C)** in Table 3b.

Q8c(v). Calculate the Deseasonalized Value labelled as **(E)** in Table 3b.

Q8c(vi). Write down the equation of the regression line used to calculate Deseasonalized Forecast values.

Q8c(vii). Calculate the Seasonalized Forecast for **2021 Quarter 3**.

Q8c(viii). Calculate the Absolute Percent Error for the forecast in the period **2020 Quarter 1**.

Q9. *Show your workings clearly and round off your answers to **three decimal places where necessary.** Use ^ symbol to represent the power notation.*

Long queues of products are frequently seen at the quality control (QC) station of a manufacturing facility.

The current QC process is carried out manually.

A time study shows that finished products arrive at the QC station according to a Poisson process at an average rate of 210 per hour. Average QC time for one product is 15 seconds.

Manual QC time follows a negative exponential distribution.

The production supervisor proposes to implement a machine to automate the QC process. The machine QC time is constant at 15 seconds for each product.

Q9a. Using Kendall's notation, identify a suitable queuing model for the:

i. Manual QC process

M/M/1; MM1

ii. Machine QC process

M/D/1; MD1

Q9b. Calculate:

i. Average arrival rate of the products to the QC station in per minute (2 marks)

ii. Average QC rate in per minute (2 marks)

iii. Utilization of the QC station. (2 marks)

Q9c. Calculate the average number of products waiting at the QC station, Lq.

Q9d. Explain in words how the implementation of machine QC process would affect the average number of products waiting at the QC station.

Q9e. The supervisor wants to implement machine QC process to cope with increased number of products arriving for QC. Suppose the product arrival rate is now 960 per hour. Assume average QC time remains at 15 seconds.

Q9e(i). What is the minimum number of machines required for the automated QC process to be in a steady state condition? Explain.

Q9e(ii). Suppose that the machine QC process has been implemented. Given the average number of products in queue, Lq is 2.22, calculate the average waiting time for a product at the QC station, Wq.

Q9e(iii). Assume that there are 5 machines at the automated QC station. Using Table 4 where n is the number of products in the system, calculate the probability that **at least 1 machine is idle.**

Table 4: Partial Workings of M/M/5 Queuing Model

n	$\frac{(\lambda/\mu)^n}{n!}$	$\frac{(\lambda/\mu)^n}{n!}P_0$ for $0 \le n \le k$	$\frac{(\lambda/\mu)^n}{k!k^{n-k}}P_0$ for $n > k$
0	1	0.013	0.338
1	4.000	0.052	0.271
2	8.000	0.104	0.216
3	10.667	0.139	0.173
4	10.667	0.139	0.139
5	8.533	0.111	0.111
6	5.689	0.074	0.089
7	3.251	0.042	0.071
8	1.625	0.021	0.057

Q10. Aishah is building a FlexSim simulation model of an assembly process with 3 operations – Operation A, Operation B and Quality Check (QC) (Figure 5a).

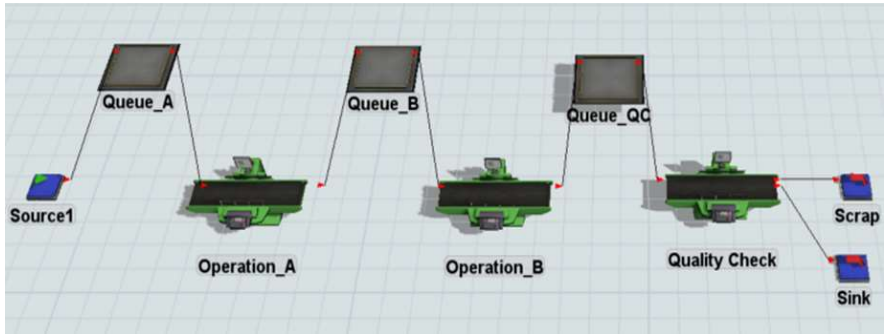

Figure 5a: Simulation Model of an Assembly Process

Q10a. Referring to Figure 5a, suppose that there is space limitation at the queue before Operation A. The queue can only hold 15 items. How do you model this in FlexSim?

Q10b. After Quality Check, items are either sent to Scrap (failed QC) or Sink (passed QC). The percentage of items passing QC is 97%. To model this, a label named 'Check' is created for each item. Items with the label value of 1 will be sent to Scrap whereas those with the label value of 2 will be sent to Sink. State the correct inputs in Source1 properties, On Creation Trigger (Figure 5b).

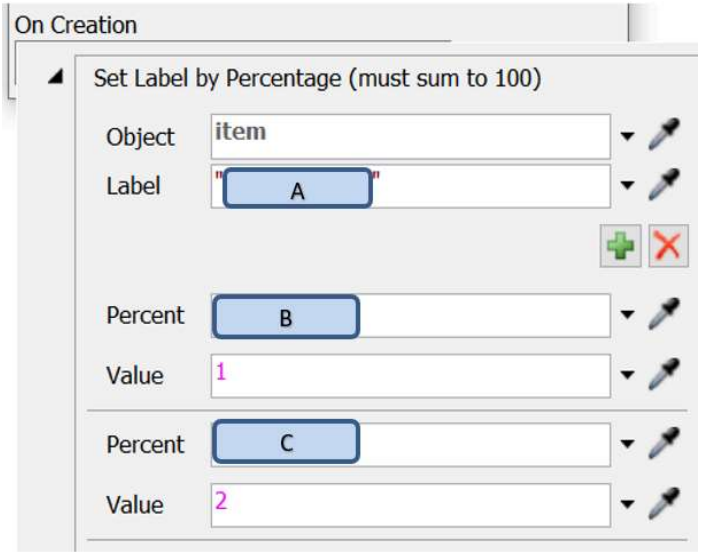

Figure 5b: On Creation Trigger of Source1

A: Check

B: 3

C: 97

Q10c. Suggest 2 ways to improve the realism of the 3D simulation model in Figure 5a.

Q10d. What are 2 pitfalls Aishah should watch out for when conducting the above simulation study?

Q11. Jordan is using FlexSim Process Flow simulation to study material handling by robots in a facility. Figure 6a shows a snapshot of the partial simulation model.

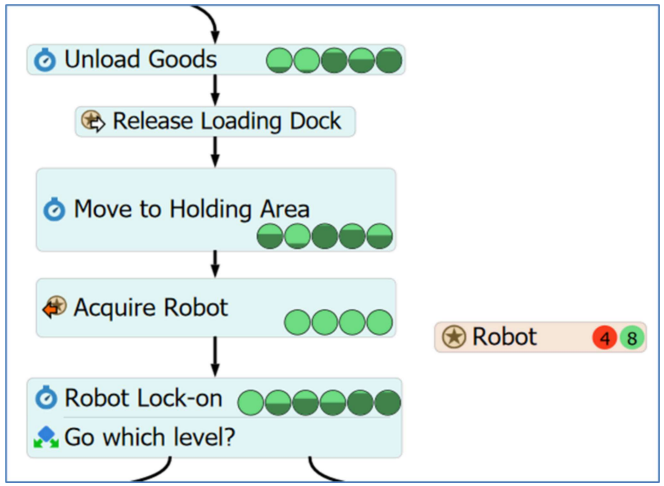

Figure 6a: FlexSim Process Flow Partial Model

Q11a. Referring to Figure 6a, state:

i. Number of items waiting for a Robot to be available = 4

ii. Number of Delay activities in the partial model = 3

Q11b. What is the function of the 'Release Loading Dock' activity?

Q11c. Figure 6b shows the statistics page of the Robot resource. Answer the following questions:

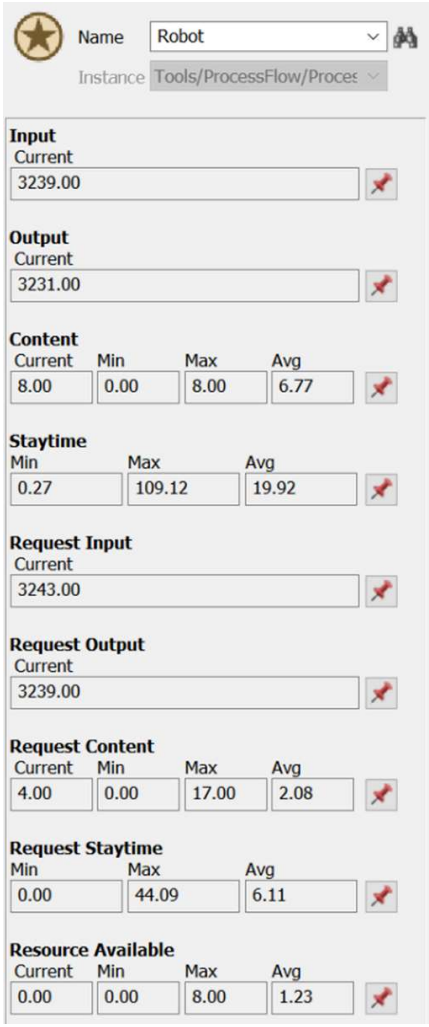

Figure 6b: Robot Resource Statistics

Q11c(i). State the average waiting time for a Robot.

6.11

Q11c(ii). How much space is to be allocated at the waiting area for Robot? Assume this should be equal to the maximum queue length.

17

Q11c(iii). Calculate the average utilization of Robot resource.

Q12. Jerin constructed the Patient Flow model in FlexSim Healthcare shown in Figure 7a to simulate a registration and consultation process in a clinic.

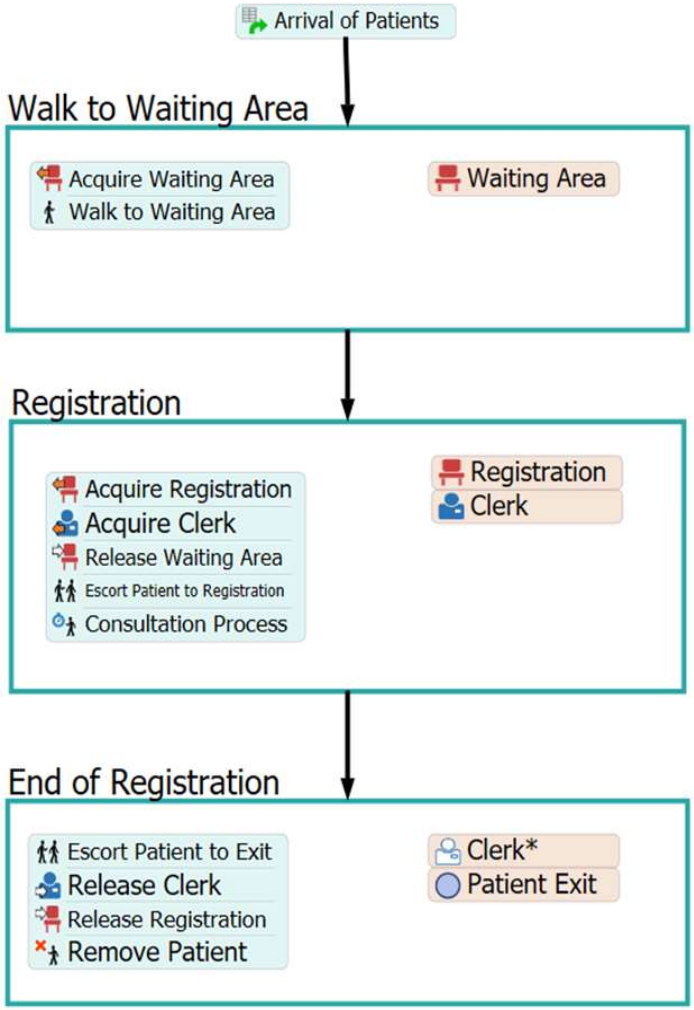

Figure 7a: Patient Flow Model in FlexSim Healthcare

Q12a. Given the patient arrival schedule below (Figure 7b), where should Jerin enter this data in the FlexSim Healthcare (HC) model?

Start (DD:)HH:MM:SS	End (DD:)HH:MM:SS	Quantity	Name
08:00:00	08:30:00	5	
08:30:00	09:30:00	3	
09:30:00	10:00:00	3	

Figure 7b: Patient Arrival Schedule

Q12b. When modeling the arrivals of patients, the options for different types of arrival spacing are shown in Figure 7c. Explain the difference between 'At scheduled time' and 'Randomly spaced'.

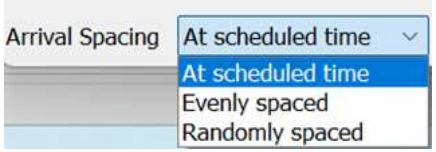

Figure 7c: Arrival Spacing Options

Q12c. Identify two roles of the Clerk according to the Patient Flow Model in Figure 7a.

Q12d. Jerin used the HC activity set 'Walk then Process' from FlexSim Healthcare library to model the 'Walk to Waiting Area' activity set in Figure 7a. What changes did he make to the 'Walk then Process' activity set?

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